An open replication of the Bank of England's structural VAR model for the UK economy*

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Abstract

I present an open-source Python replication of the eight-variable Bayesian structural VAR that the Bank of England uses to disentangle domestic and global drivers of the UK business cycle (Brignone and Piffer, 2025). The model combines a Minnesota normal-inverse-Wishart prior with zero and sign restrictions imposed via the algorithm of Arias et al. (2018), identifying world demand, world energy, world supply, UK demand, UK supply and UK monetary policy shocks, with two unidentified residual shocks. The replication serves as the baseline and conditioning member of PolicyEngine Macro: it produces forecasts with credible bands, structural readings of the latest shocks, and forecast-revision narratives, but does not score reforms. Validation against the published paper shows that zero restrictions hold exactly on every accepted identification draw, sign restrictions hold on every draw and on the median impulse responses, the unrestricted oil-price response to a world supply shock is positive (+6.06 on impact), and the identified global shocks explain 47.4 per cent of UK GDP and 42.4 per cent of UK CPI forecast-error variance at the one-year horizon, against roughly 40 and 50 per cent in the paper — inside deliberately wide calibration bands. Internal consistency holds to numerical precision: variance decompositions sum to one within 10^{-10} and the historical decomposition reconstructs the data within 10^{-8}. An honest out-of-sample look from the 2024Q2 data edge shows the model's GDP forecast (year-on-year medians of 1.3–2.0 per cent) bracketing the ONS outturns, while CPI inflation (medians of 2.3–2.8 per cent) undershoots the 2025 re-acceleration to 3.8 per cent, an error shared by the Bank's own August 2024 projections. The model is served through hosted MCP tools, making a central-bank-grade structural VAR callable from any AI workflow.

Keywords: structural VAR, sign restrictions, Bayesian estimation, UK economy, replication, open source

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1 Introduction

In July 2025 the Bank of England published, for the first time, a full technical description of the Bayesian structural vector autoregression (SVAR) it uses in the monetary policy round to separate domestic from global drivers of the UK business cycle (Brignone and Piffer, 2025). The model is small — eight quarterly variables, six identified structural shocks — but it carries substantial weight in the Bank's process: it supplies impulse responses, forecast-error variance decompositions, real-time structural shock readings, and a round-by-round structural narrative for revisions to the Monetary Policy Report forecast. No official replication package accompanies the paper.

This paper documents an independent, open-source Python replication of that model, developed as part of *PolicyEngine Macro*, PolicyEngine's suite of open macroeconomic models for the UK and US (Ghenis and PolicyEngine contributors, 2025).[1] Within the suite the SVAR plays a specific role: it is the *baseline and conditioning member*. It reads the current state of the economy in structural-shock terms, produces unconditional forecasts with credible bands, and generates revision narratives that other models in the suite condition on. It deliberately does *not* score policy reforms — reform scoring is the province of the suite's overlapping-generations, macroeconometric and microsimulation members — because a sign-restricted SVAR identifies shocks, not policy rules.

The contribution is threefold. First, *replication as a public good*: the Bank released no code or data, and its two world aggregates are internal, unpublished series. Reconstructing the pipeline — data assembly, Minnesota-prior Bayesian estimation, zero-and-sign-restriction identification following Arias et al. (2018), and the standard outputs — makes the Bank's analytical machinery inspectable and extensible by anyone. Second, *validation with numbers rather than vibes*: Section 5 compares the replication against the paper's published figures quantitatively, reporting exact satisfaction of every identifying restriction, the sign of key unrestricted responses, and variance-decomposition shares side by side with the paper's, together with machine-precision internal-consistency checks. Third, *operationalisation*: the model runs as a set of hosted Model Context Protocol (MCP) tools — `forecast_uk`, `latest_shocks`, `model_summary` — so that forecasts and structural readings are callable from AI agents and ordinary scripts alike.

Section 2 sets out the model, priors, identification scheme and output mathematics. Section 3 identifies the source paper and delimits the replication scope. Section 4 describes the implementation. Section 5 reports the calibration and validation results. Section 6 shows a current forecast from the 2024Q2 data edge and confronts it with subsequent ONS outturns and the Bank's own contemporaneous projections. Section 7 concludes with limitations.

2 The model

2.1 Reduced form

Let y_t be the $k \times 1$ vector ($k = 8$) of quarterly variables listed in Table 1. The reduced form is a VAR(p) in (mostly log) levels with a constant and six Covid-19 dummies:

$$y_t \;=\; c \;+\; \sum_{l=1}^{p} \Pi_l \, y_{t-l} \;+\; \sum_{j=1}^{6} \delta_j \, d_{j,t} \;+\; u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

[1] Code: https://github.com/PolicyEngine/boe-var-model; hosted interface: https://macromod.vercel.app/svar/.

Table 1: Variables, transformations and ordering (source paper Table 1)

#	Variable	Enters model as	Shown in figures as
1	Real world GDP (UK-trade-weighted)	$100 \cdot \log$	YoY growth
2	World CPI (UK-trade-weighted)	$100 \cdot \log$	YoY growth
3	Real oil price in sterling	$100 \cdot \log$	YoY growth
4	Bank Rate	level (%)	level
5	Sterling effective exchange rate (ERI)	$100 \cdot \log$	level
6	UK CPI, seasonally adjusted (CPISA)	$100 \cdot \log$	YoY growth
7	UK CPI Energy	$100 \cdot \log$	YoY growth
8	Real UK GDP	$100 \cdot \log$	YoY growth

where each dummy $d_{j,t}$ equals one in exactly one of the quarters 2020Q1–2021Q2 and zero otherwise, absorbing the extreme pandemic observations in the spirit of the pandemic prior of Cascaldi-Garcia (2022). The source paper does not state the lag length; the replication uses $p = 4$ as a configurable default.

Reduced-form innovations map to orthonormal structural shocks through an impact matrix B:

$$u_t = B\,\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, I_k), \qquad \Sigma = BB'. \tag{2}$$

B is parameterised through the Cholesky factor of Σ and an orthogonal rotation:

$$B = \mathrm{chol}(\Sigma)\,Q, \qquad Q'Q = I_k. \tag{3}$$

Any orthogonal Q delivers the same reduced-form fit; identification consists of restricting the admissible set of Q.

2.2 Prior and posterior

Estimation is Bayesian in the tradition of Doan et al. (1984) and Litterman (1986), implemented as in Giannone et al. (2015): a Minnesota normal-inverse-Wishart (NIW) prior on (Π, Σ) centred on independent random walks, combined with a sum-of-coefficients prior that disciplines the deterministic component. The conjugate posterior is standard: Σ is drawn from an inverse Wishart and the stacked coefficients Π from a matrix normal conditional on Σ. The replication uses overall tightness $\lambda = 0.2$ and sum-of-coefficients weight $\mu = 1$ (the paper does not report its hyperparameters), with the Covid dummies entering as exogenous regressors rather than through the full dummy-observation algebra of Cascaldi-Garcia (2022).

2.3 Identification: zero and sign restrictions

Six of the eight shocks are identified — world demand (WD), world energy (WE), world supply (WS), UK demand (UD), UK supply (US) and UK monetary policy (MP) — while two are left unidentified, one global (U1) and one UK-specific (U2), to absorb residual volatility. All restrictions apply to the impact matrix B only. Table 2 reproduces the restriction matrix (Table 2 of the source paper) exactly as encoded in the replication.

The economic content is conventional: a positive world demand shock raises world and UK output and prices; an expansionary (disinflationary) world energy shock raises real activity while lowering oil prices, world CPI and UK CPI including its energy component; a positive world

Table 2: Identifying restrictions on impact (source paper Table 2)

Variable	WD	WE	WS	U1	UD	US	MP	U2
World GDP	+	+	+		0	0	0	0
World CPI	+	−	−		0	0	0	0
Oil price		−			0	0	0	0
Bank Rate					+		+	
Exchange rate								
UK CPISA	+	−	−		+	−	−	
CPI Energy	+	−	0					
Real UK GDP	+	+	+		+	+	−	

Notes: Blank cells are unrestricted. The zero block on the three world variables encodes the small-open-economy assumption: UK shocks (UD, US, MP) and the UK unidentified shock (U2) have no contemporaneous impact on world variables. The global unidentified shock U1 is exempt from that block — that exemption is what makes it "global". The zero on CPI Energy under WS exogenises energy prices with respect to the world supply shock, separating it from the world energy shock.

supply shock raises output and lowers prices with a zero energy-price impact; a UK demand shock raises UK GDP, CPI and Bank Rate; a UK supply shock raises GDP and lowers CPI; and a monetary tightening raises Bank Rate while lowering CPI and GDP.

For each posterior draw (Π, Σ), Q is drawn with the algorithm of Arias et al. (2018) so that the *zero* restrictions hold exactly by construction: with $L = \mathrm{chol}(\Sigma)$, column q_j is a standard normal vector projected onto the null space of the zero restrictions on Lq_j and of the previously drawn columns, then normalised. The *sign* restrictions are then checked on $B = LQ$, flipping a column's (unidentified) sign where doing so satisfies its restrictions, and accepting or rejecting the whole draw as in Uhlig (2005) and Rubio-Ramírez et al. (2010). Following the source paper's use of the invariance results of Chan et al. (2025), column permutations are also searched before rejecting; the replication conservatively permutes only within groups of shocks that share the same zero-restriction pattern, since with zero restrictions the Haar measure is permutation-invariant only within such groups.

2.4 Outputs

Write (1) in companion form with companion matrix F and selection matrix J so that $\Psi_h = JF^hJ'$ is the reduced-form moving-average coefficient at horizon h. The structural impulse response of variable i to shock j at horizon h is

$$\mathrm{IRF}_{ij}(h) = \left[\Psi_h B\right]_{ij}, \tag{4}$$

reported as pointwise posterior medians with 68 and 90 per cent credible bands across accepted (Π, Σ, Q) triples. The forecast-error variance decomposition (FEVD) at horizon H is

$$\mathrm{FEVD}_{ij}(H) \;=\; \frac{\sum_{h=0}^{H}\left[\Psi_h B\right]_{ij}^2}{\sum_{h=0}^{H}\left[\Psi_h \Sigma \Psi_h'\right]_{ii}}, \qquad \sum_{j=1}^{k} \mathrm{FEVD}_{ij}(H) = 1. \tag{5}$$

Estimated structural shocks are recovered as $\hat{\varepsilon}_t = B^{-1}\hat{u}_t$ at each accepted draw. The historical decomposition splits each variable's path into the cumulative contribution of each structural shock, the Covid-dummy component and the deterministic component:

$$y_{i,t} = \underbrace{\sum_{j=1}^{k} \sum_{h=0}^{t-1} \left[\Psi_h B\right]_{ij} \hat{\varepsilon}_{j,t-h}}_{\text{structural shocks}} + \underbrace{y_{i,t}^{\text{covid}}}_{\text{dummies}} + \underbrace{y_{i,t}^{\text{det}}}_{\text{initial conditions, constant}}, \tag{6}$$

an additive identity that must reconstruct the data exactly — a property Section 5 verifies to 10^{-8}.

3 The source paper and replication scope

The model replicated here is documented in Brignone and Piffer (2025): Davide Brignone and Michele Piffer, *A structural VAR model for the UK economy*, Bank of England Macro Technical Paper No. 3, published 18 July 2025. It is one of the first papers in the Bank's Macro Technical Paper series, which documents the modelling toolkit supporting the Monetary Policy Committee, and it describes the SVAR used on a round-by-round basis to provide a structural narrative for revisions to the Monetary Policy Report forecast. A companion paper by the same authors develops the forecast-revision methodology in general form (Brignone and Piffer, 2026).

Replication scope. The replication targets the paper's core pipeline and its standard outputs (the paper's Figures 2–6): Bayesian estimation, identification, impulse responses, FEVDs, estimated structural shocks and historical decompositions, plus the Section-5 forecast-revision machinery in reduced scope. The estimation sample matches the paper exactly: **1992Q1–2023Q2** (126 quarters), beginning with UK inflation targeting and ending a year before the data edge because recent quarters are revision-prone. Data through **2024Q2** are retained for the forecasting and forecast-revision exercises, matching the vintage of the Bank's August 2024 Monetary Policy Report round (Bank of England, 2024).

Approximated internal series. Two of the eight variables — UK-trade-weighted world GDP and world CPI — are internal Bank series that are not published. The replication proxies them with trade-weighted aggregates constructed from OECD and IMF sources, and constructs the real sterling oil price from Brent crude converted to sterling and deflated by UK CPI, since the paper does not fully specify its construction. UK series (Bank Rate, sterling ERI, CPISA, CPI Energy, real GDP) come from the Bank of England and the Office for National Statistics (Office for National Statistics, 2026, 2025a). Because the world aggregates are proxies and the Bank released no replication package, the global block can be validated only against the published figures, not against the underlying data — a caveat the quantitative comparison in Section 5 makes explicit.

Documented deviations. Beyond the proxy world series, the replication documents its deviations openly: an assumed lag length of $p = 4$ (unstated in the paper), default rather than hierarchically optimised prior hyperparameters, Covid dummies as exogenous regressors rather than the full pandemic-prior algebra, permutation search restricted to zero-pattern-equivalent groups, and current data vintages in place of the August 2024 vintages. Each deviation is a configurable parameter or a flagged approximation in the codebase, not a silent choice.

4 Method and implementation

4.1 Code architecture

The replication is a small, dependency-light Python package (`boe_var`, src layout; NumPy, SciPy, pandas, matplotlib) with a module per stage of the pipeline:

- `data.py` — `load_data()` returns the transformed quarterly panel (a `PeriodIndex` DataFrame in the Table 1 ordering, $100 \cdot \log$ except Bank Rate), assembled by a separate download script from public sources;
- `bvar.py` — the `BVAR` class: NIW posterior sampling of (Π, Σ) with Minnesota and sum-of-coefficients priors and exogenous Covid dummies, plus companion-form and residual helpers;
- `identification.py` — the Table 2 restrictions as machine-readable constants (`ZERO_RESTRICTIONS`, `SIGN_RESTRICTIONS`), the Arias et al. (2018) `draw_Q` routine, and `identify()`, which pairs posterior draws with accepted rotations;
- `analysis.py` — IRFs, FEVDs, estimated shocks, historical decompositions, forecast bands and the composite (forecast-revision) impulse response, with plotting helpers.

An end-to-end script reproduces the paper's Figures 2–6 and writes a numerical summary; the test suite (Section 5) runs the same code on the real dataset with a fixed seed.

4.2 Estimation and identification runs

The production configuration draws **10,000** posterior draws, of which **751** pass the sign-restriction accept/reject step (a 7.5 per cent acceptance rate), with $p = 4$ lags and hyperparameters $\lambda = 0.2$, $\mu = 1$. Zero restrictions are satisfied by construction on every draw; sign restrictions are enforced by rejection, so they too hold exactly on every accepted draw. Because the Arias et al. (2018) zero-restriction sampler does not draw Q uniformly from the restricted set, the slow validation path additionally computes the algorithm's importance weights (each requiring a numerical Jacobian per accepted draw); the importance-weighted effective sample size is **350.3**, and weighted and unweighted results are compared in Section 5. The forecast-revision exercise re-estimates through 2024Q2 with 6,000 draws (487 accepted, weighted ESS 207.5).

4.3 Hosted MCP tools

Within PolicyEngine Macro the model is wrapped by an integration layer — a `pe-macro` command-line interface and a Model Context Protocol (MCP) server, deployed automatically to Modal on every merge — exposing three tools:

- `forecast_uk` — unconditional forecasts of the eight variables from the data edge, with median paths and credible bands, in year-on-year units for growth variables;
- `latest_shocks` — the posterior sign probabilities and magnitudes of the six identified structural shocks in the most recent quarter, i.e. a structural reading of "what just hit the economy";

- `model_summary` — the model card: specification, sample, restriction matrix, acceptance statistics and validation status.

The same tools back the model page at `https://macromod.vercel.app/svar/`. This makes the SVAR usable as the suite's conditioning member: agents and scripts can pull a structural baseline before invoking the reform-scoring models.

5 Calibration and validation

Validation is against ground truth — the restrictions and published figures of Brignone and Piffer (2025) — rather than the mere absence of exceptions. The checks fall into three tiers: exact invariants that must hold on every draw, sign checks on quantities the identification does *not* restrict, and quantitative comparisons with the paper's reported magnitudes inside deliberately wide bands that respect the replication's proxy data and Monte-Carlo noise. All checks run in continuous integration on the real dataset.

Exact invariants. On *every* accepted identification draw: (i) each element of B restricted to zero in Table 2 is zero to machine precision (largest violation below 10^{-9}) — the Arias et al. (2018) construction guarantees this and the test suite verifies it; (ii) every sign restriction holds strictly ($s \cdot B_{vj} > 0$), the accept/reject invariant; (iii) $BB' = \Sigma$ within 10^{-7}, so every accepted B is a valid structural factorisation. Median impact impulse responses also respect every Table 2 sign.

Unrestricted sign checks. The paper's key overidentifying check is that the real oil price — whose response to a world supply shock is left *unrestricted* — rises after a positive world supply shock. The replication reproduces this: the median impact response of the sterling real oil price to a one-standard-deviation world supply shock is +6.06 ($100 \cdot \log$ units), positive as in the paper's Figure 2.

Comparison with published magnitudes. Table 3 collects the model-versus-paper comparison. The headline calibration target is the share of UK forecast-error variance attributed to the identified global shocks (world demand + energy + supply) at the one-year horizon: the paper reports roughly 40 per cent for UK GDP and roughly 50 per cent for UK CPI. The replication's CI validation configuration yields **47.4** and **42.4** per cent respectively; the full 10,000-draw production run yields 40.9 and 50.1 per cent. Both sit within the deliberately wide $[30, 60]$ per cent acceptance band, which is chosen to flag gross regressions without being flaky to Monte-Carlo variation and to the proxy world aggregates. The importance-weighted slow benchmark (ESS 350.3) moves the shares towards the paper's values, consistent with the weights correcting the Arias et al. (2018) sampler towards the uniform-over-rotations posterior.

Internal consistency. Identities are verified to numerical precision rather than assumed: FEVD shares sum to one at every horizon on every draw within 10^{-10} (equation 5); the historical decomposition (6) — structural shocks plus Covid component plus deterministic component — reconstructs the observed data within 10^{-8}, and the stochastic component equals the sum over shocks within 10^{-10}; and in the forecast-revision exercise the identity "forecast at T = pseudo-forecast at $T-1$ + composite impulse response" holds per draw with a maximum absolute error of 5.3×10^{-12} across all draws, horizons and variables.

Table 3: Model versus paper: validation summary

Check	Replication	Paper	Verdict
Zero restrictions (Table 2)	exact, every draw ($< 10^{-9}$)	imposed	pass
Sign restrictions (Table 2)	every draw and median IRF	imposed	pass
$BB' = \Sigma$	every draw ($< 10^{-7}$)	by construction	pass
Oil price after world supply shock (impact)	+6.06 (unrestricted)	positive (Fig. 2)	pass
Global share, UK GDP FEVD, 1 yr	47.4% (40.9% at 10k draws)	∼40%	in band
Global share, UK CPI FEVD, 1 yr	42.4% (50.1% at 10k draws)	∼50%	in band
FEVD rows sum to one	$< 10^{-10}$, all horizons/draws	identity	pass
Historical decomposition reconstructs data	$< 10^{-8}$	identity	pass
Forecast-revision adding-up identity	$< 5.3 \times 10^{-12}$	identity	pass
Shock narrative (2008, 2022–23 episodes)	reproduced	Fig. 5	pass
Largest domestic CPI contributor	UK supply	monetary policy	disclosed miss

Notes: "In band" refers to the $[30, 60]\%$ calibration band around the paper's approximate published shares. Production run: 10,000 posterior draws, 751 accepted, importance-weighted ESS 350.3. The final row is a known, documented discrepancy attributed to the proxy world aggregates and the smaller accepted sample; see the repository's sense-check log.

Narrative reproduction. The estimated shock series reproduce the paper's episodes: the 2008 collapse in world demand, the contractionary world energy shocks of 2022, UK monetary loosening readings after 2022 followed by tightening readings in 2023, and muted shocks during 2020–21 by construction of the Covid dummies. Known caveats are equally documented: median UK GDP responses to world demand and supply shocks decay through zero at long horizons where the paper's remain positive (the paper's path lies inside the replication's bands), and within the global supply side the energy shock contributes more than the non-energy supply shock to world GDP and oil-price variance, where the paper's figure suggests the reverse ordering; the paper's claims about the *combined* supply side are matched.

6 Results: a forecast from the 2024Q2 data edge

6.1 The forecast

With the model estimated through 2024Q2 — the same vintage the Bank faced in its August 2024 round — the unconditional forecast puts year-on-year UK GDP growth medians between **1.3 and 2.0 per cent** and year-on-year CPI inflation medians between **2.3 and 2.8 per cent** over the forecast horizon, with credible bands that widen with horizon as posterior parameter and shock uncertainty compound. The structural reading at the data edge (the `latest_shocks` output) assigns a 0.81 posterior probability to a positive world demand shock and a 0.80 probability to a contractionary (price-raising) world energy shock in 2024Q2, with a 0.75 probability of a positive UK supply shock — a configuration in which global demand supports activity while the energy and domestic supply readings pull inflation in opposite directions.

6.2 An honest out-of-sample look

Table 4 confronts the forecast with what actually happened, using ONS outturns and the Bank's contemporaneous August 2024 Monetary Policy Report projections (Bank of England, 2024).

The scorecard is mixed in an informative way. On *activity*, the model does well: ONS

Table 4: Forecast from 2024Q2 versus outturns and the Bank's August 2024 projections

	SVAR replication (median, YoY)	Outturn (ONS)	BoE MPR, Aug. 2024
GDP growth, 2024 (annual)	≈ 1.3%	1.1%	$\approx 1\frac{1}{4}\%$
GDP growth, 2025 (annual)	1.3–2.0%	1.3%	≈ 1%
CPI inflation, end-2024	2.3–2.8%	2.5% (Dec.)	$\approx 2\frac{3}{4}\%$
CPI inflation, 2025	2.3–2.8%	rising to 3.8% (Sep. peak)	falling towards 2%

Notes: Outturns from ONS quarterly national accounts and consumer price inflation releases (Office for National Statistics, 2026, 2025a,b): annual GDP growth of 1.1% in 2024 and 1.3% in 2025; CPI inflation of 2.5% in the twelve months to December 2024, rising through 2025 to a peak of 3.8% (September 2025) before easing to 3.2% by November 2025. Bank projections are the August 2024 Monetary Policy Report's modal paths, shown at comparable points. SVAR bands widen with horizon; the 2025 CPI outturn sits outside the median range but within the upper credible band.

outturns of 1.1 per cent growth in 2024 and 1.3 per cent in 2025 sit at, or just below, the bottom of the median range, and comfortably inside the bands — the model correctly read an economy growing modestly rather than re-accelerating or stalling. On *inflation*, the model's 2.3–2.8 per cent median range captures end-2024 exactly (2.5 per cent in December 2024) but misses the 2025 re-acceleration, in which CPI inflation climbed to 3.8 per cent by September 2025 on energy, administered prices and persistent services inflation before easing. This is an honest failure to report — and a shared one: the Bank's own August 2024 projections had inflation falling back towards target over 2025, so the miss reflects genuinely unforecastable 2024–25 cost developments (and, for the replication, a data edge frozen at 2024Q2) rather than a defect peculiar to the replication. The widening credible bands do their job: the 2025 outturn lies within the upper band even as it escapes the median range.

The forecast-revision machinery adds the structural narrative: relative to a pseudo-forecast made one quarter earlier, the 2024Q2 shock readings revise the year-on-year GDP path up by a peak of +0.40 percentage points and the CPI path down by −0.22 percentage points on impact — the model's structural account of why the data edge changed the outlook, exactly the round-by-round use case the Bank built the model for.

7 Limitations and conclusion

Three limitations should frame any use of the model, and each is disclosed in the codebase as well as here.

The data edge is frozen at 2024Q2. The estimation sample ends in 2023Q2 and the conditioning data in 2024Q2, matching the source paper's vintage. Forecasts and shock readings therefore describe the economy as of mid-2024; the out-of-sample comparison in Section 6 is possible precisely because the edge is frozen, but operational use requires re-running the data pipeline on current vintages, and results will shift with revisions.

Internal Bank series are approximated. The UK-trade-weighted world GDP and world CPI series are unpublished; the replication's OECD/IMF-based proxies, the assumed lag length, the default prior hyperparameters and the simplified pandemic-prior treatment mean the global block matches the paper qualitatively and within bands, not exactly. The one documented substantive discrepancy — UK monetary policy not being the largest domestic contributor to CPI variance — most likely traces to these approximations.

This is not an official Bank product. The replication is an independent open-source reconstruction from the published paper. It is not endorsed by, and does not speak for, the Bank of England; where the replication and the paper disagree, the paper is ground truth. Nor does the model score reforms: within PolicyEngine Macro it is the baseline and conditioning member, and reform scoring belongs to the suite's other models.

Within those limits, the exercise shows that a central bank's operational structural VAR can be rebuilt in the open: every identifying restriction verified on every draw, published magnitudes matched within stated bands, identities holding to machine precision, honest out-of-sample performance on the record, and the whole apparatus callable through hosted tools. Replication of policy-institution models, with numbers attached, is both feasible and useful — and this paper is offered as a template for doing it.

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